Spatial Rank-Based Augmentation for Nonparametric Online Monitoring and Adaptive Sampling of Big Data Streams

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S1.1: Property 1.1 and proof

In this section, we will propose and prove Property 1.1 of the SRAS algorithm in Section 3.2.

Property 1.1: Assume all elements in the bandwidth vector $\boldsymbol{H}_j^{\mathcal{O}(t)} \to \infty$ for any $\boldsymbol{H}_j^{\mathcal{O}(t)} \in \mathcal{H}$. Then for any unobserved data stream $j \in \mathcal{U}(t)$, we have that $\hat{X}_{t,j} \to \frac{1}{m_0}\sum_{i=-m_0+1}^{0} X_{i,j}$. Further, if $m_0 \to \infty$, $\hat{X}_{t,j} \to \mu_{0,j}$.

Remark: Property 1.1 shows that with a large bandwidth value, the augmentation for the unobservable data stream will equal to its IC sample mean based on historical data, which shows the robustness of the augmentation method. Further, if the number of samples $m_0 \to \infty$, the underlying IC distribution is assumed to be completely known (Zou et al. 2012) and therefore the augmentation would equal to its IC mean, thus free of dependence of historical observations.

Proof of Property 1.1:

According to equation (3), for any data stream $j \in \mathcal{U}(t)$, we have that

$$\hat{X}_{t,j} = \hat{\mathbb{E}}\left[X_{t,j}\middle|\boldsymbol{X}_t^{\mathcal{O}}\right] = \frac{\sum_{\tau=-m_0+1}^{0} K_{\boldsymbol{H}_j^{\mathcal{O}(t)}}\left(\boldsymbol{X}_t^{\mathcal{O}} - \boldsymbol{X}_\tau^{\mathcal{O}}\right)X_{\tau,j}}{\sum_{i=-m_0+1}^{0} K_{\boldsymbol{H}_j^{\mathcal{O}(t)}}\left(\boldsymbol{X}_t^{\mathcal{O}} - \boldsymbol{X}_i^{\mathcal{O}}\right)} = \sum_{\tau=-m_0+1}^{0} \omega_{\tau,j} X_{\tau,j},$$

where $\omega_{\tau,j} = \frac{K_{\boldsymbol{H}_j^{\mathcal{O}(t)}}\left(\boldsymbol{X}_t^{\mathcal{O}} - \boldsymbol{X}_\tau^{\mathcal{O}}\right)}{\sum_{i=-m_0+1}^{0} K_{\boldsymbol{H}_j^{\mathcal{O}(t)}}\left(\boldsymbol{X}_t^{\mathcal{O}} - \boldsymbol{X}_i^{\mathcal{O}}\right)}$ denotes the weight for each sample $X_{\tau,j}$ and $\sum_{\tau=-m_0+1}^{0} \omega_{\tau,j} = 1$.

Since any element in the bandwidth vector $\boldsymbol{H}_j^{\mathcal{O}(t)} \to \infty$, for $\forall \tau \in \{-m_0 + 1, \cdots, 0\}$, we have

$$\lim_{\boldsymbol{H}_j^{\mathcal{O}(t)} \to \infty} K_{\boldsymbol{H}_j^{\mathcal{O}(t)}}\left(\boldsymbol{X}_t^{\mathcal{O}} - \boldsymbol{X}_\tau^{\mathcal{O}}\right) = K(0), \forall \tau.$$

Therefore, we can find that $\omega_{\tau_1,j} = \omega_{\tau_2,j}, \forall \tau_1, \tau_2 \in \{-m_0 + 1, \cdots, 0\}$. Combining with the equation such that $\sum_{\tau=-m_0+1}^{0} \omega_{\tau,j} = 1$, as any element in the bandwidth vector $\boldsymbol{H}_j^{\mathcal{O}} \to \infty, \forall \boldsymbol{H}_j^{\mathcal{O}} \in \mathcal{H}$, we can derive that

$$\omega_{\tau,j} \to \frac{1}{m_0}, \forall \tau.$$

Accordingly, we can obtain that

$$\hat{X}_{t,j} = \sum_{\tau=-m_0+1}^{0} \omega_{\tau,j} X_{\tau,j} \to \frac{1}{m_0} \sum_{i=-m_0+1}^{0} X_{i,j}.$$

Furthermore, as $m_0 \to \infty$, we have that $\frac{1}{m_0}\sum_{i=-m_0+1}^{0} X_{i,j} \to \mu_{0,j}$, where $\mu_{0,j}$ is the IC mean for data stream j. In other words, as $m_0 \to \infty$, $\hat{X}_{t,j} \to \mu_{0,j}, j \in \mathcal{U}(t)$. To this end, we have proved this property.

S1.2: Property 1.2 and proof

In this section, we will propose and prove Property 1.2 of the SRAS algorithm in Section 3.2.

Property 1.2: Assume all elements in the candidate bandwidth vector $\boldsymbol{H}_j^{\mathcal{O}(t)}$ are equal, i.e., $\boldsymbol{H}_j^{\mathcal{O}(t)} = h_j^{\mathcal{O}(t)}\mathbf{1}_r$, and define kernel density function $f_j\left(\boldsymbol{X}_t^{\mathcal{O}}\right)$ such that $f_j\left(\boldsymbol{X}_t^{\mathcal{O}}\right) > 0$, whose estimate is $\hat{f}_j\left(\boldsymbol{X}_t^{\mathcal{O}}\right) = \frac{1}{m_0}\sum_{i=-m_0+1}^{0} K_{h_j^{\mathcal{O}(t)}}\left(\boldsymbol{X}_t^{\mathcal{O}} - \boldsymbol{X}_i^{\mathcal{O}}\right)$. Let $\left\{h_{m_0}^{j,\mathcal{O}(t)}\right\}$ be a priori chosen sequence of bandwidth parameters for $h_j^{\mathcal{O}(t)}$, satisfying $\lim_{m_0 \to \infty} h_{m_0}^{j,\mathcal{O}(t)} = 0$ and $\lim_{m_0 \to \infty} m_0 \left(h_{m_0}^{j,\mathcal{O}(t)}\right)^r = \infty$. Assume $\sup_{\boldsymbol{X}_t^{\mathcal{O}}} f_j\left(\boldsymbol{X}_t^{\mathcal{O}}\right) < \infty$, $\sup_{\boldsymbol{X}_t^{\mathcal{O}}} \left|\mathbb{E}\left[X_{t,j} \middle| \boldsymbol{X}_t^{\mathcal{O}}\right]\right| f_j\left(\boldsymbol{X}_t^{\mathcal{O}}\right) < \infty$ and $\sup_{\boldsymbol{X}_t^{\mathcal{O}}} \sigma_j^2\left(\boldsymbol{X}_t^{\mathcal{O}}\right) f_j\left(\boldsymbol{X}_t^{\mathcal{O}}\right) < \infty$. Then under the null hypothesis, as $m_0 \to \infty$, $\hat{X}_{t,j} = \hat{\mathbb{E}}\left[X_{t,j} \middle| \boldsymbol{X}_t^{\mathcal{O}}\right] \xrightarrow{P} \mathbb{E}\left[X_{t,j} \middle| \boldsymbol{X}_t^{\mathcal{O}}\right]$, where $\xrightarrow{P}$ denotes converging in probability.

Remark: Property 1.2 shows that the NW estimator is weakly consistent, which assures the unbiasedness $\hat{X}_{t,j}$ for augmentation (Bierens 1994).

Proof of Property 1.2:

To prove this property, it is sufficient to show that

$$\hat{f}_j(\boldsymbol{X}_t^{\mathcal{O}}) \xrightarrow{P} f_j(\boldsymbol{X}_t^{\mathcal{O}}), \tag{12}$$

$$\hat{X}_{t,j}\hat{f}_j(\boldsymbol{X}_t^{\mathcal{O}}) \xrightarrow{P} \mathbb{E}\left[X_{t,j}\middle|\boldsymbol{X}_t^{\mathcal{O}}\right] f_j(\boldsymbol{X}_t^{\mathcal{O}}). \tag{13}$$

We begin with the proof of equation (12). Given the assumptions such that $\lim_{m_0\to\infty} h_{m_0}^{j,\mathcal{O}(t)} = 0$, $\lim_{m_0\to\infty} m_0\left(h_{m_0}^{j,\mathcal{O}(t)}\right)^r = \infty$, and $\sup_{\boldsymbol{X}_t^{\mathcal{O}}} f_j(\boldsymbol{X}_t^{\mathcal{O}}) < \infty$, we first derive the asymptotic unbiasedness from

$$\mathbb{E}\left[\hat{f}_j(\boldsymbol{X}_t^{\mathcal{O}})\right] = \int \left(h_{m_0}^{j,\mathcal{O}(t)}\right)^{-r} K\left(\frac{\boldsymbol{X}_t^{\mathcal{O}} - \boldsymbol{X}^{\mathcal{O}}}{h_{m_0}^{j,\mathcal{O}(t)}}\right) f_j(\boldsymbol{X}^{\mathcal{O}}) d\boldsymbol{X}^{\mathcal{O}} = \int f_j\left(\boldsymbol{X}_t^{\mathcal{O}} - h_{m_0}^{j,\mathcal{O}(t)}\boldsymbol{X}^{\mathcal{O}}\right) K(\boldsymbol{X}^{\mathcal{O}}) d\boldsymbol{X}^{\mathcal{O}}$$

$$\to f_j(\boldsymbol{X}_t^{\mathcal{O}}) \int K(\boldsymbol{X}^{\mathcal{O}}) d\boldsymbol{X}^{\mathcal{O}} = f_j(\boldsymbol{X}_t^{\mathcal{O}})$$

by bounded convergence. Then the variance vanishes at order $\mathrm{O}\left(1/m_0\left(h_{m_0}^{j,\mathcal{O}(t)}\right)^r\right)$, as

$$m_0\left(h_{m_0}^{j,\mathcal{O}(t)}\right)^r \mathrm{Var}\left[\hat{f}_j(\boldsymbol{X}_t^{\mathcal{O}})\right] = m_0\left(h_{m_0}^{j,\mathcal{O}(t)}\right)^r \frac{1}{m_0^2} \sum_{i=-m_0+1}^{0} \mathrm{Var}\left[\left(h_{m_0}^{j,\mathcal{O}(t)}\right)^{-r} K\left(\frac{\boldsymbol{X}_t^{\mathcal{O}} - \boldsymbol{X}_i^{\mathcal{O}}}{h_{m_0}^{j,\mathcal{O}(t)}}\right)\right]$$

$$= \mathbb{E}\left[\left(h_{m_0}^{j,\mathcal{O}(t)}\right)^{-r}\left(K\left(\frac{\boldsymbol{X}_t^{\mathcal{O}} - \boldsymbol{X}_i^{\mathcal{O}}}{h_{m_0}^{j,\mathcal{O}(t)}}\right)\right)^2\right] - \left(h_{m_0}^{j,\mathcal{O}(t)}\right)^r\left(\mathbb{E}\left[\left(h_{m_0}^{j,\mathcal{O}(t)}\right)^{-r} K\left(\frac{\boldsymbol{X}_t^{\mathcal{O}} - \boldsymbol{X}_i^{\mathcal{O}}}{h_{m_0}^{j,\mathcal{O}(t)}}\right)\right]\right)^2$$

$$= \int f_j\left(\boldsymbol{X}_t^{\mathcal{O}} - h_{m_0}^{j,\mathcal{O}(t)}\boldsymbol{X}^{\mathcal{O}}\right) K^2(\boldsymbol{X}^{\mathcal{O}}) d\boldsymbol{X}^{\mathcal{O}}$$

$$- \left(h_{m_0}^{j,\mathcal{O}(t)}\right)^r \left(\int f_j\left(\boldsymbol{X}_t^{\mathcal{O}} - h_{m_0}^{j,\mathcal{O}(t)}\boldsymbol{X}^{\mathcal{O}}\right) K(\boldsymbol{X}^{\mathcal{O}}) d\boldsymbol{X}^{\mathcal{O}}\right)^2 \to f_j(\boldsymbol{X}_t^{\mathcal{O}}) \int K^2(\boldsymbol{X}^{\mathcal{O}}) d\boldsymbol{X}^{\mathcal{O}}$$

by bounded convergence. Thus, the equation (12) has been proved.

Let $\sigma_j^2(\boldsymbol{X}_t^{\mathcal{O}}) = \mathbb{E}\left[X_{i,j}^2\middle|\boldsymbol{X}_i^{\mathcal{O}} = \boldsymbol{X}_t^{\mathcal{O}}\right]$, for $f_j(\boldsymbol{X}_t^{\mathcal{O}}) > 0$. Similarly, for the proof of equation (13), under all the given assumptions, it follows that

$$\mathbb{E}\left[\hat{X}_{t,j}\hat{f}_j\left(\boldsymbol{X}_t^{\mathcal{O}}\right)\right] = \mathbb{E}\left[X_{i,j}\left(h_{m_0}^{j,\mathcal{O}(t)}\right)^{-r} K\left(\frac{\boldsymbol{X}_t^{\mathcal{O}} - \boldsymbol{X}_i^{\mathcal{O}}}{h_{m_0}^{j,\mathcal{O}(t)}}\right)\right] = \mathbb{E}\left[\mathbb{E}\left[X_{i,j}\middle|\boldsymbol{X}_i^{\mathcal{O}}\right]\left(h_{m_0}^{j,\mathcal{O}(t)}\right)^{-r} K\left(\frac{\boldsymbol{X}_t^{\mathcal{O}} - \boldsymbol{X}_i^{\mathcal{O}}}{h_{m_0}^{j,\mathcal{O}(t)}}\right)\right]$$

$$= \int \left(h_{m_0}^{j,\mathcal{O}(t)}\right)^{-r} K\left(\frac{\boldsymbol{X}_t^{\mathcal{O}} - \boldsymbol{X}^{\mathcal{O}}}{h_{m_0}^{j,\mathcal{O}(t)}}\right)\mathbb{E}\left[X_{t,j}\middle|\boldsymbol{X}^{\mathcal{O}}\right] f_j\left(\boldsymbol{X}^{\mathcal{O}}\right) d\boldsymbol{X}^{\mathcal{O}}$$

$$= \int \mathbb{E}\left[X_{t,j}\middle|\boldsymbol{X}_t^{\mathcal{O}} - h_{m_0}^{j,\mathcal{O}(t)}\boldsymbol{X}^{\mathcal{O}}\right] f_j\left(\boldsymbol{X}_t^{\mathcal{O}} - h_{m_0}^{j,\mathcal{O}(t)}\boldsymbol{X}^{\mathcal{O}}\right) K\left(\boldsymbol{X}^{\mathcal{O}}\right) d\boldsymbol{X}^{\mathcal{O}} \to \mathbb{E}\left[X_{t,j}\middle|\boldsymbol{X}_t^{\mathcal{O}}\right] f_j\left(\boldsymbol{X}_t^{\mathcal{O}}\right)$$

by bounded convergence, and

$$m_0\left(h_{m_0}^{j,\mathcal{O}(t)}\right)^r \mathrm{Var}\left[\hat{X}_{t,j}\hat{f}_j\left(\boldsymbol{X}_t^{\mathcal{O}}\right)\right] = \left(h_{m_0}^{j,\mathcal{O}(t)}\right)^{-r}\frac{1}{m_0}\sum_{i=-m_0+1}^{0}\mathrm{Var}\left[X_{i,j}K\left(\frac{\boldsymbol{X}_t^{\mathcal{O}} - \boldsymbol{X}_i^{\mathcal{O}}}{h_{m_0}^{j,\mathcal{O}(t)}}\right)\right]$$

$$= \mathbb{E}\left[X_{i,j}^2\left(h_{m_0}^{j,\mathcal{O}(t)}\right)^{-r}\left(K\left(\frac{\boldsymbol{X}_t^{\mathcal{O}} - \boldsymbol{X}_i^{\mathcal{O}}}{h_{m_0}^{j,\mathcal{O}(t)}}\right)\right)^2\right]$$

$$-\left(h_{m_0}^{j,\mathcal{O}(t)}\right)^r\left(\mathbb{E}\left[X_{i,j}\left(h_{m_0}^{j,\mathcal{O}(t)}\right)^{-r}K\left(\frac{\boldsymbol{X}_t^{\mathcal{O}} - \boldsymbol{X}_i^{\mathcal{O}}}{h_{m_0}^{j,\mathcal{O}(t)}}\right)\right]\right)^2$$

$$= \int \left(h_{m_0}^{j,\mathcal{O}(t)}\right)^{-r} K\left(\frac{\boldsymbol{X}_t^{\mathcal{O}} - \boldsymbol{X}^{\mathcal{O}}}{h_{m_0}^{j,\mathcal{O}(t)}}\right)\sigma_j^2\left(\boldsymbol{X}^{\mathcal{O}}\right) f_j\left(\boldsymbol{X}^{\mathcal{O}}\right) d\boldsymbol{X}^{\mathcal{O}} - \left(h_{m_0}^{j,\mathcal{O}(t)}\right)^r\left(\mathbb{E}\left[\hat{X}_{t,j}\hat{f}_j\left(\boldsymbol{X}_t^{\mathcal{O}}\right)\right]\right)^2$$

$$= \int \sigma_j^2\left(\boldsymbol{X}_t^{\mathcal{O}} - h_{m_0}^{j,\mathcal{O}(t)}\boldsymbol{X}^{\mathcal{O}}\right) f_j\left(\boldsymbol{X}_t^{\mathcal{O}} - h_{m_0}^{j,\mathcal{O}(t)}\boldsymbol{X}^{\mathcal{O}}\right) K^2\left(\boldsymbol{X}^{\mathcal{O}}\right) d\boldsymbol{X}^{\mathcal{O}}$$

$$-\left(h_{m_0}^{j,\mathcal{O}(t)}\right)^r\left(\mathbb{E}\left[\hat{X}_{t,j}f_j\left(\boldsymbol{X}_t^{\mathcal{O}}\right)\right]\right)^2 \to \sigma_j^2\left(\boldsymbol{X}_t^{\mathcal{O}}\right) f_j\left(\boldsymbol{X}_t^{\mathcal{O}}\right)\int K^2\left(\boldsymbol{X}^{\mathcal{O}}\right) d\boldsymbol{X}^{\mathcal{O}}$$

by bounded convergence. Thus, we have proved equation (13). Based on equations (12) and (13), we can accordingly derive that $\hat{X}_{t,j} = \widehat{\mathbb{E}}\left[X_{t,j}\middle|\boldsymbol{X}_t^{\mathcal{O}}\right] \xrightarrow{P} \mathbb{E}\left[X_{t,j}\middle|\boldsymbol{X}_t^{\mathcal{O}}\right]$. Therefore, we have completed the proof.

S1.3: Property 2 and proof

In this section, we will propose and prove Property 2 of the SRAS algorithm in Section 3.2.

Property 2: Assume all elements in the bandwidth vector $\boldsymbol{H}_j^{\mathcal{O}} \to \infty$ for any $\boldsymbol{H}_j^{\mathcal{O}} \in \mathcal{H}$ and $m_0 \to \infty$. Under regularity conditions, assume there exists a mean shift $\boldsymbol{\delta}$ occurring to $\boldsymbol{X}_t$, where the observable subset is denoted as $\boldsymbol{\delta}^{\mathcal{O}}$. Let $\boldsymbol{R}_t^0$ denote the IC augmented spatial rank $\boldsymbol{R}_t^0 = \mathbb{E}_{\boldsymbol{X}}\left[U\left(\boldsymbol{V}^0(t)\right)\right]$, where

$\boldsymbol{V}^0(t) = \boldsymbol{M}_0\left(\widetilde{\boldsymbol{X}}_t - \boldsymbol{X}\right)$ when $\boldsymbol{\delta}^{\mathcal{O}} = \boldsymbol{0}$. With a general nonsingular covariance matrix $\boldsymbol{\Sigma}_0$ such that $\boldsymbol{M}_0 = [\boldsymbol{M}_0^{\mathcal{O}},\ \boldsymbol{M}_0^{\mathcal{U}}]$, where $\boldsymbol{M}_0^{\mathcal{O}}$ and $\boldsymbol{M}_0^{\mathcal{U}}$ denote the submatrices of $\boldsymbol{M}_0$ containing observable and unobservable columns respectively, then the OC augmented spatial rank $\boldsymbol{R}_t^* = \mathbb{E}_{\boldsymbol{X}}\left[U\left(\boldsymbol{V}^0(t) + \boldsymbol{M}_0^{\mathcal{O}}\boldsymbol{\delta}^{\mathcal{O}}\right)\right]$. Specifically, $\boldsymbol{R}_t^* = \boldsymbol{R}_t^0$ if and only if $\boldsymbol{\delta}^{\mathcal{O}} = \boldsymbol{0}$.

Remark: Property 2 guarantees the differentiation performance of the augmented spatial rank. If there are one or more OC data streams that are observed, the corresponding augmented spatial rank will change accordingly, which ensures the detection efficiency of the SRAS algorithm.

Proof of Property 2:

Given that $m_0 \to \infty$, the underlying distribution is completely known with IC mean vector $\boldsymbol{\mu}_0$ and IC covariance matrix $\boldsymbol{\Sigma}_0$, thus the transformation matrix $\boldsymbol{M}_0$ such that $\boldsymbol{M}_0 = \boldsymbol{\Sigma}_0^{-\frac{1}{2}}$ is also known as stated. According to equation (4), we have

$$\boldsymbol{R}_t = \lim_{m_0\to\infty}\frac{1}{m_0}\sum_{h=-m_0+1}^{0} U\left(\boldsymbol{M}_0\left(\widetilde{\boldsymbol{X}}_t - \boldsymbol{X}_h\right)\right) = \mathbb{E}_{\boldsymbol{X}}\left[U\left(\boldsymbol{M}_0\left(\widetilde{\boldsymbol{X}}_t - \boldsymbol{X}\right)\right)\right] = \mathbb{E}_{\boldsymbol{X}}\left[U\left(\boldsymbol{V}(t)\right)\right],$$

where $\boldsymbol{V}(t) = \boldsymbol{M}_0\left(\widetilde{\boldsymbol{X}}_t - \boldsymbol{X}\right)$.

Without loss of generality, assume that $\mathcal{O}(t) = \{1, \cdots, r\}$ (we can always reorder $\widetilde{\boldsymbol{X}}_t$ to get such a format). Given that any candidate bandwidth vector $\boldsymbol{H}_j^{\mathcal{O}} \to \infty, \forall \boldsymbol{H}_j^{\mathcal{O}} \in \mathcal{H}$, according to Property 1.1, the NW estimation of the unobservable data streams $\hat{X}_{t,j} \to \mu_{0,j}, j \in \mathcal{U}(t)$ and thus $\widetilde{\boldsymbol{X}}_t$ can be defined as

$$\widetilde{\boldsymbol{X}}_t = \begin{pmatrix} \boldsymbol{X}_t^{\mathcal{O}} \\ \boldsymbol{\mu}_0^{\mathcal{U}} \end{pmatrix},$$

where $\boldsymbol{X}_t^{\mathcal{O}}$ is the r-dimensional observed subset of $\boldsymbol{X}_t$ and $\boldsymbol{\mu}_0^{\mathcal{U}}$ is the $(p-r)$-dimensional unobservable subset of $\boldsymbol{\mu}_0$. Assume that the data streams follow the linear model $\boldsymbol{X}_t = \boldsymbol{\mu}_0 + \boldsymbol{\Omega}_0\boldsymbol{\varepsilon}_t$, where $\boldsymbol{\Omega}_0 = \boldsymbol{M}_0^{-1} = \boldsymbol{\Sigma}_0^{\frac{1}{2}}$ and $\boldsymbol{\varepsilon}_t$ is the standardized and independent residual such that $\mathrm{cov}(\boldsymbol{\varepsilon}_t) = \boldsymbol{I}_p$ and $\mathbb{E}[\boldsymbol{\varepsilon}_t] = \boldsymbol{0}$. When there exists a mean shift $\boldsymbol{\delta}$ such that current observation $\boldsymbol{X}_t^* = \boldsymbol{\mu}_0 + \boldsymbol{\delta} + \boldsymbol{\Omega}_0\boldsymbol{\varepsilon}_t$, we can then derive that

$$\widetilde{\boldsymbol{X}}_t = \begin{pmatrix} \boldsymbol{\mu}_0^{\mathcal{O}} + \boldsymbol{\delta}^{\mathcal{O}} + (\boldsymbol{\Omega}_0\boldsymbol{\varepsilon}_t)^{\mathcal{O}} \\ \boldsymbol{\mu}_0^{\mathcal{U}} \end{pmatrix},$$

where $\boldsymbol{\mu}_0^{\mathcal{O}}$, $\boldsymbol{\delta}^{\mathcal{O}}$ and $(\boldsymbol{\Omega}_0\boldsymbol{\varepsilon}_t)^{\mathcal{O}}$ are the r-dimensional observed subset of $\boldsymbol{\mu}_0$, $\boldsymbol{\delta}$ and $\boldsymbol{\Omega}_0\boldsymbol{\varepsilon}_t$, respectively. Therefore, we can derive that

$$\widetilde{\boldsymbol{X}}_t - \boldsymbol{X} = \begin{pmatrix} \boldsymbol{X}_t^{\mathcal{O}} \\ \boldsymbol{\mu}_0^{\mathcal{U}} \end{pmatrix} - \boldsymbol{X} = \begin{pmatrix} \boldsymbol{\mu}_0^{\mathcal{O}} + \boldsymbol{\delta}^{\mathcal{O}} + (\boldsymbol{\Omega}_0\boldsymbol{\varepsilon}_t)^{\mathcal{O}} \\ \boldsymbol{\mu}_0^{\mathcal{U}} \end{pmatrix} - (\boldsymbol{\mu}_0 + \boldsymbol{\Omega}_0\boldsymbol{\varepsilon}) = \begin{pmatrix} \boldsymbol{\delta}^{\mathcal{O}} + (\boldsymbol{\Omega}_0\boldsymbol{\varepsilon}_t)^{\mathcal{O}} \\ \mathbf{0}^{\mathcal{U}} \end{pmatrix} - \boldsymbol{\Omega}_0\boldsymbol{\varepsilon},$$

$$\boldsymbol{V}(t) = \boldsymbol{M}_0\big(\widetilde{\boldsymbol{X}}_t - \boldsymbol{X}\big) = \boldsymbol{M}_0\big(\widetilde{\boldsymbol{X}}_t - \boldsymbol{X}\big) = \boldsymbol{M}_0 \begin{pmatrix} \boldsymbol{\delta}^{\mathcal{O}} + (\boldsymbol{\Omega}_0\boldsymbol{\varepsilon}_t)^{\mathcal{O}} \\ \mathbf{0}^{\mathcal{U}} \end{pmatrix} - \boldsymbol{\varepsilon}.$$

Accordingly, we can derive that

$$\boldsymbol{R}_t = \mathbb{E}_{\boldsymbol{\varepsilon}}\big[U\big(\boldsymbol{V}(t)\big)\big] = \mathbb{E}_{\boldsymbol{\varepsilon}}\left[U\left(\boldsymbol{M}_0 \begin{pmatrix} \boldsymbol{\delta}^{\mathcal{O}} + (\boldsymbol{\Omega}_0\boldsymbol{\varepsilon}_t)^{\mathcal{O}} \\ \mathbf{0}^{\mathcal{U}} \end{pmatrix} - \boldsymbol{\varepsilon}\right)\right].$$

With a general nonsingular covariance matrix $\boldsymbol{\Sigma}_0$ such that $\boldsymbol{\Omega}_0 = \{w_{i,j}\}, \boldsymbol{M}_0 = \{m_{i,j}\}$, we have

$$\boldsymbol{M}_0 \begin{pmatrix} \boldsymbol{\delta}^{\mathcal{O}} + (\boldsymbol{\Omega}_0\boldsymbol{\varepsilon}_t)^{\mathcal{O}} \\ \mathbf{0}^{\mathcal{U}} \end{pmatrix} = \boldsymbol{M}_0 \begin{pmatrix} \sum_{z\in\mathcal{P}} w_{1,z}\varepsilon_{t,z} + \delta_1 \\ \vdots \\ \sum_{z\in\mathcal{P}} w_{r,z}\varepsilon_{t,z} + \delta_r \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} \sum_{y\in\mathcal{O}(t)} m_{1,y} \sum_{z\in\mathcal{P}} w_{y,z}\varepsilon_{t,z} + \sum_{y\in\mathcal{O}(t)} m_{1,y}\delta_y \\ \vdots \\ \sum_{y\in\mathcal{O}(t)} m_{p,y} \sum_{z\in\mathcal{P}} w_{y,z}\varepsilon_{t,z} + \sum_{y\in\mathcal{O}(t)} m_{p,y}\delta_y \end{pmatrix}$$

$$= \begin{pmatrix} \sum_{z\in\mathcal{P}} \varepsilon_{t,z} \left(\sum_{y\in\mathcal{O}(t)} m_{1,y}w_{y,z} \right) + \sum_{y\in\mathcal{O}(t)} m_{1,y}\delta_y \\ \vdots \\ \sum_{z\in\mathcal{P}} \varepsilon_{t,z} \left(\sum_{y\in\mathcal{O}(t)} m_{p,y}w_{y,z} \right) + \sum_{y\in\mathcal{O}(t)} m_{p,y}\delta_y \end{pmatrix}.$$

Note that $\boldsymbol{\delta}^{\mathcal{O}} = \mathbf{0}$ represents the state when there is no mean shift in the observed data streams, and thus denote the IC $\boldsymbol{V}^0(t) = \boldsymbol{M}_0 \begin{pmatrix} (\boldsymbol{\Omega}_0\boldsymbol{\varepsilon}_t)^{\mathcal{O}} \\ \mathbf{0}^{\mathcal{U}} \end{pmatrix} - \boldsymbol{\varepsilon}$ and the OC $\boldsymbol{V}^*(t) = \boldsymbol{M}_0 \begin{pmatrix} \boldsymbol{\delta}^{\mathcal{O}} + (\boldsymbol{\Omega}_0\boldsymbol{\varepsilon}_t)^{\mathcal{O}} \\ \mathbf{0}^{\mathcal{U}} \end{pmatrix} - \boldsymbol{\varepsilon}$ such that $\boldsymbol{\delta}^{\mathcal{O}} \neq \mathbf{0}$. Accordingly, we derive that

$$\boldsymbol{V}^*(t) = \boldsymbol{V}^0(t) + \boldsymbol{M}_0^{\mathcal{O}}\boldsymbol{\delta}^{\mathcal{O}},$$

where $\boldsymbol{M}_0^{\mathcal{O}}$ and $\boldsymbol{M}_0^{\mathcal{U}}$ denote the submatrices of $\boldsymbol{M}_0$ containing observable and unobservable column vectors respectively such that $\boldsymbol{M}_0 = [\boldsymbol{M}_0^{\mathcal{O}}, \quad \boldsymbol{M}_0^{\mathcal{U}}]$. Thus, we derive the IC $\boldsymbol{R}_t^0$ and OC $\boldsymbol{R}_t^*$ given by

$$\boldsymbol{R}_t^0 = \mathbb{E}_{\boldsymbol{\varepsilon}}\big[U\big(\boldsymbol{V}^0(t)\big)\big],$$

$$\boldsymbol{R}_t^* = \mathbb{E}_{\boldsymbol{\varepsilon}}\big[U\big(\boldsymbol{V}^*(t)\big)\big] = \mathbb{E}_{\boldsymbol{\varepsilon}}\big[U\big(\boldsymbol{V}^0(t) + \boldsymbol{M}_0^{\mathcal{O}}\boldsymbol{\delta}^{\mathcal{O}}\big)\big].$$

If $\boldsymbol{M}_0^{\mathcal{O}}\boldsymbol{\delta}^{\mathcal{O}} \neq \mathbf{0}$, it should be noted that we are always able to rotate the coordinate axis to let $\boldsymbol{M}_0^{\mathcal{O}}\boldsymbol{\delta}^{\mathcal{O}}$ be in the direction $(1,0,\cdots,0)'$. Therefore, without loss of generality, suppose $\boldsymbol{M}_0^{\mathcal{O}}\boldsymbol{\delta}^{\mathcal{O}} = (1,0,\cdots,0)'$, then the first element in the direction $U\big(\boldsymbol{V}^0(t) + \boldsymbol{M}_0^{\mathcal{O}}\boldsymbol{\delta}^{\mathcal{O}}\big)$ should always be no less than the first element in the direction $U\big(\boldsymbol{V}^0(t)\big)$ with only finite equalities, and thus $\boldsymbol{R}_t^* \neq \boldsymbol{R}_t^0$. To this end, to prove Property 2, it suffices to show that $\boldsymbol{M}_0^{\mathcal{O}}\boldsymbol{\delta}^{\mathcal{O}} = \mathbf{0}$ if and only if $\boldsymbol{\delta}^{\mathcal{O}} = \mathbf{0}$, which is equivalent to show that there is no non-zero solution in the linear system $\boldsymbol{M}_0^{\mathcal{O}}\boldsymbol{\delta}^{\mathcal{O}} = \mathbf{0}$ (in $\boldsymbol{\delta}^{\mathcal{O}}$). This statement is obvious since $\boldsymbol{M}_0 = \boldsymbol{\Sigma}_0^{-\frac{1}{2}}$ is nonsingular, which leads to the conclusion that all the column vectors in the observed subset $\boldsymbol{M}_0^{\mathcal{O}}$ are linearly independent, i.e., $\boldsymbol{M}_0^{\mathcal{O}}$ is full column rank. Therefore, the linear system $\boldsymbol{M}_0^{\mathcal{O}}\boldsymbol{\delta}^{\mathcal{O}} = \mathbf{0}$ (in $\boldsymbol{\delta}^{\mathcal{O}}$) only has a zero solution such that $\boldsymbol{\delta}^{\mathcal{O}} = \mathbf{0}$. Accordingly, when there is any mean shift occurring in the observed data streams such that $\boldsymbol{\delta}^{\mathcal{O}} \neq \mathbf{0}$, the corresponding $\boldsymbol{R}_t^*$ indicates a change accordingly such that $\boldsymbol{R}_t^* \neq \boldsymbol{R}_t^0$. Thus, we have proved this property.

S1.4: Proof of Property 3.1

In this section, we will prove Property 3.1 of the SRAS algorithm in Section 3.2.

Property 3.1: Assume that in the Aug(1) method $\Delta^{(t)} = \Delta > \mathbb{E}\big[R_{t,i}^2 \big| i \in \mathcal{O}(t), i \in \mathcal{I}(t)\big] - \mathbb{E}\big[R_{t,j}^2 \big| j \in \mathcal{U}(t)\big]$, the control limit $K \to \infty$, $m_0 \to \infty$, and all elements in $\boldsymbol{H}_j^{\mathcal{O}} \to \infty$ for any $\boldsymbol{H}_j^{\mathcal{O}} \in \mathcal{H}$, where $\mathcal{I}(t)$ represents the set of IC data streams at time t. Suppose all the data streams are exchangeable and let U_t denote the set of data streams $i \in \mathcal{P}$ which are never observed till some finite time t, i.e., $U_t = \bigcap_{\tau=1}^{t} \mathcal{U}(\tau)$. As time $t \to \infty$, $\mathbb{P}(U_t = \emptyset) \to 1$, where $\emptyset$ represents the empty set, $\mathbb{P}(\cdot)$ represents the probability of an event.

To prove Property 3.1, we consider the following lemma first.

Lemma 1: Assume that in the Aug(1) method $\Delta^{(t)} = \Delta > \mathbb{E}\big[R_{t,i}^2 \big| i \in \mathcal{O}(t), i \in \mathcal{I}(t)\big] - \mathbb{E}\big[R_{t,j}^2 \big| j \in \mathcal{U}(t)\big]$, the control limit $K \to \infty$, $m_0 \to \infty$ and all elements in the bandwidth vector $\boldsymbol{H}_j^{\mathcal{O}} \to \infty$ for any $\boldsymbol{H}_j^{\mathcal{O}} \in \mathcal{H}$, where $\mathcal{I}(t)$ represents the set of IC data streams at time t. Suppose all the data streams

are exchangeable and let U_t denote the set of data streams $i \in \mathcal{P}$ which are never observed till some finite time t, i.e., $U_t = \bigcap_{\tau=1}^{t} \mathcal{U}(\tau)$. Suppose there exists a data stream $j \in U_t = \bigcap_{\tau=1}^{t} \mathcal{U}(\tau)$, there is nonzero probability that the data stream j will be observed at next time $(t+1)$, i.e., $\mathbb{P}(j \in \mathcal{O}(t+1) | j \in U_t) > 0$ for any t.

Remark: Lemma 1 shows that there is always nonzero probability of a data stream that has never been observed to be observed next time.

Proof of Lemma 1:

Recall that $j_{(1)}, \cdots, j_{(p)}$ are the data stream indices such that $z_{t,j_{(1)}} \geq z_{t,j_{(2)}} \geq \cdots \geq z_{t,j_{(p)}}$. Suppose $j \in U_t = \bigcap_{\tau=1}^{t} \mathcal{U}(\tau)$, and thus we can derive the following through induction:

$$\mathbb{P}(j \in \mathcal{O}(t+1) | j \in U_t) \propto \mathbb{P}\left(z_{t,j} - z_{t,j_{(r)}} \geq 0\right) = \mathbb{P}\left(\sum_{\tau=1}^{t} (1-\lambda)^{t-\tau} \lambda \left(\tilde{L}_{\tau,j} - \tilde{L}_{\tau,j_{(r)}}\right) \geq 0\right) =$$

$$= \mathbb{P}\left(\sum_{\tau=1}^{t} (1-\lambda)^{t-\tau} \lambda \left(\Delta + R_{\tau,j}^2 - R_{\tau,j_{(r)}}^2\right) \geq 0\right).$$

Let $A_{\tau,j} = \Delta + R_{\tau,j}^2 - R_{\tau,j_{(r)}}^2$ and $G_{t,j} = \sum_{\tau=1}^{t} (1-\lambda)^{t-\tau} \lambda A_{\tau j}$. For $\tau = 1, \cdots, t$, $A_{\tau,j}$ are i.i.d. with mean

$$\mathbb{E}\left[A_{\tau,j}\right] = \Delta + \mathbb{E}\left[R_{\tau,j}^2 \middle| j \in \mathcal{U}(t)\right] - \mathbb{E}\left[R_{\tau,j_{(r)}}^2\right] > 0.$$

Therefore, as $t \to \infty$, $G_{t,j} \xrightarrow{D} G_j$ with mean $\mathbb{E}\left[G_j\right] > 0$, where $\xrightarrow{D}$ denotes converging in distribution. It further leads to the conclusion that for any $t = 1,2, \cdots$,

$$\mathbb{P}\left(z_{t,j} - z_{t,j_{(r)}} \geq 0\right) > 0.$$

This means that $\mathbb{P}(j \in \mathcal{O}(t+1) | j \in U_t) > 0$ for any t. Thus, we have proved lemma 1.

Proof of Property 3.1:

Based on the Bayes' theorem, we can derive the following for any $j \in \mathcal{P}$ at time t recursively:

$$\mathbb{P}(j \in U_t) = \mathbb{P}(j \in \mathcal{U}(t) | j \in U_{t-1}) \mathbb{P}(j \in U_{t-1}) = \left(1 - \mathbb{P}(j \in \mathcal{O}(t) | j \in U_{t-1})\right) \mathbb{P}(j \in U_{t-1})$$

$$= \prod_{\tau=2}^{t} \left(1 - \mathbb{P}(j \in \mathcal{O}(\tau) | j \in U_{\tau-1})\right) \mathbb{P}\left(j \in \mathcal{U}(1)\right).$$

According to lemma 1, $\mathbb{P}(j \in \mathcal{O}(\tau) | j \in U_{\tau-1}) > 0$ for any τ. Let $\varepsilon = \min_{\tau} \mathbb{P}(j \in \mathcal{O}(\tau) | j \in U_{\tau-1}) > 0$, and thus we derive that for any $\tau = 2, \cdots, t$,

$$1 - \mathbb{P}(j \in \mathcal{O}(\tau) | j \in U_{\tau-1}) \leq 1 - \varepsilon.$$

As a result, we can derive that

$$\mathbb{P}(j \in U_t) \leq (1-\varepsilon)^{t-1} \mathbb{P}\big(j \in \mathcal{U}(1)\big),$$

where $0 < 1 - \varepsilon < 1$ and $0 < \mathbb{P}\big(j \in \mathcal{U}(1)\big) < 1$. Thus, as time $t \to \infty$, $(1-\varepsilon)^{t-1} \to 0$. This means that as time $t \to \infty$, $\mathbb{P}(j \in U_t) \to 0$ for any $j \in \mathcal{P}$. It further indicates that none of the data stream j can be unobserved as $t \to \infty$, which leads to the conclusion that $\mathbb{P}(U_t = \emptyset) \to 1$. Therefore, we have proved property 3.1.

Corollary 1: Assume that in the Aug(1) method $\lambda = 1$, $\Delta^{(t)} = \Delta$, the control limit $K \to \infty$, $m_0 \to \infty$ and all elements in the bandwidth vector $\boldsymbol{H}_j^{\mathcal{O}} \to \infty$ for any $\boldsymbol{H}_j^{\mathcal{O}} \in \mathcal{H}$, where $\mathcal{I}(t)$ represents the set of IC data streams at time t. Suppose all the data streams are exchangeable and let U_t denote the set of data streams $i \in \mathcal{P}$ which are never observed till some finite time t, i.e., $U_t = \bigcap_{\tau=1}^{t} \mathcal{U}(\tau)$. Suppose there exists a data stream $j \in U_t = \bigcap_{\tau=1}^{t} \mathcal{U}(\tau)$ and let $P_{t+1,j}(\Delta)$ denote the probability that the data stream j will be observed at next time $(t+1)$, i.e., $\mathbb{P}(j \in \mathcal{O}(t+1) | j \in U_t)$, then $P_{t+1,j}(\Delta_2) \geq P_{t+1,j}(\Delta_1)$ for any time t if $\Delta_2 > \Delta_1$.

Remark: Corollary 1 further shows that larger value of Δ leads to no smaller probability of one unobserved data stream to be observed at the next time under regularity conditions. Combining these facts, it is reasonable to conclude that it is more likely for all data streams to be observed at least once at a finite time with a higher value of Δ.

Proof of Corollary 1:

Similar to lemma 1, as $\lambda = 1$, we derive the following for any time t:

$$P_{t+1,j}(\Delta_1) \propto \mathbb{P}\left(z_{t,j} - z_{t,j_{(r)}} \geq 0 \middle| \Delta_1\right) = \mathbb{P}\left(\Delta_1 + \left(R_{t,j}^{(1)}\right)^2 - \left(R_{t,j_{(r)}}^{(1)}\right)^2 \geq 0\right),$$

$$P_{t+1,j}(\Delta_2) \propto \mathbb{P}\left(z_{t,j} - z_{t,j_{(r)}} \geq 0 \middle| \Delta_2\right) = \mathbb{P}\left(\Delta_2 + \left(R_{t,j}^{(2)}\right)^2 - \left(R_{t,j_{(r)}}^{(2)}\right)^2 \geq 0\right),$$

where $\Delta_2 > \Delta_1$ and the superscript "$^{(k)}$" represent relevant quantities for Aug(1) method with value Δ_k. Thus, we derive that

$$P_{t+1,j}(\Delta_2) - P_{t+1,j}(\Delta_1) \propto \mathbb{P}\left(\Delta_2 + \left(R_{t,j}^{(2)}\right)^2 - \left(R_{t,j_{(r)}}^{(2)}\right)^2 \geq 0\right) - \mathbb{P}\left(\Delta_1 + \left(R_{t,j}^{(1)}\right)^2 - \left(R_{t,j_{(r)}}^{(1)}\right)^2 \geq 0\right).$$

We claim that $R_{t,j}^{(1)}$ and $R_{t,j}^{(2)}$ have the same distributions and so do $R_{t,j_{(r)}}^{(1)}$ and $R_{t,j_{(r)}}^{(2)}$ (the proof is almost the same as Lemma 2.1, and thus is omitted here), thus leads to the result such that

$$\Delta_2 + \left(R_{t,j}^{(2)}\right)^2 - \left(R_{t,j_{(r)}}^{(2)}\right)^2 > \Delta_1 + \left(R_{t,j}^{(1)}\right)^2 - \left(R_{t,j_{(r)}}^{(1)}\right)^2.$$

Therefore, $\mathbb{P}\left(\Delta_2 + \left(R_{t,j}^{(2)}\right)^2 - \left(R_{t,j_{(r)}}^{(2)}\right)^2 \geq 0\right) \geq \mathbb{P}\left(\Delta_1 + \left(R_{t,j}^{(1)}\right)^2 - \left(R_{t,j_{(r)}}^{(1)}\right)^2 \geq 0\right)$, which further leads to a conclusion that for any time $t = 1,2,\cdots$,

$$P_{t+1,j}(\Delta_2) \geq P_{t+1,j}(\Delta_1)$$

if $\Delta_2 > \Delta_1$. Thus, we have proved corollary 1.

S1.5: Proof of Property 3.2

In this section, we will prove Property 3.2 of the SRAS algorithm in Section 3.2.

Property 3.2: Assume that in the Aug(1) method $\Delta^{(t)} = \Delta$, $m_0 \to \infty$, all elements in $\boldsymbol{H}_j^{\mathcal{O}} \to \infty$ for any $\boldsymbol{H}_j^{\mathcal{O}} \in \mathcal{H}$, and suppose that after time t_0, there is a shift such that $\mathbb{E}(\boldsymbol{X}_t) \neq \boldsymbol{\mu}_0$. Let $\mathcal{V}$ be the set of OC data streams such that $\mathcal{V} = \left\{j: \mathbb{E}\left[R_{t,j}^2 | j \in \mathcal{O}(t)\right] > \mathbb{E}\left[R_{t,i}^2 | i \in \mathcal{U}(t)\right] + \Delta\right\}$. Then once the data stream $j^* \in \mathcal{V}$ is observed at time t, there is a nonzero probability that this data stream j^* will be kept observed forever, i.e., $j^* \in \mathcal{O}(\tau)$ for any $\tau \geq t$.

Proof of Property 3.2:

Consider that at time $t > t_0$, the OC data stream X_{t,j^*} is observable (i.e., $j^* \in \mathcal{O}(t)$). Then $z_{t,j^*} > z_{t,i}$ for any data stream $i \in \mathcal{U}(t)$. Recall that $j_{(1)}, \cdots, j_{(p)}$ are the data stream indices such that $z_{t,j_{(1)}} \geq z_{t,j_{(2)}} \geq \cdots \geq z_{t,j_{(p)}}$. Define the difference between the increments of augmented squared spatial rank statistics on data stream j^* and $j_{(r+1)}$ at time $t+n$ as

$$D_{n,j^*} = \tilde{L}_{t+n,j^*} - \tilde{L}_{t+n,j_{(r+1)}} = \tilde{L}_{t+n,j^*} - R_{t+n,j_{(r+1)}}^2 - \Delta.$$

Let $K_{N,j^*} = \sum_{n=1}^{N}(1-\lambda)^{N-n}\lambda D_{n,j^*}$ and $\tau_K = \inf_N\left\{K_{N,j^*} \leq 0\right\}$. From the definition of τ_K, we can see that $K_{N,j^*} > 0$ for any $N < \tau_K$, which further indicates that $z_{t+N,j^*} > z_{t+N,j_{(r+1)}}$ for any $N < \tau_K$. In other words, j^* is always observed at and before time τ_K. As a result, D_{n,j^*} for $n = 1, \cdots, \tau_K - 1$ are i.i.d. with mean

$$\mathbb{E}\left[D_{n,j^*}\right] = \mathbb{E}\left[R^2_{t,j^*} \middle| j^* \in \mathcal{O}(t)\right] - \mathbb{E}\left[R^2_{t,j_{(r+1)}}\right] - \Delta > 0.$$

Therefore, as $N \to \infty$, $K_{N,j^*} \xrightarrow{D} K_{j^*}$ with mean $\mathbb{E}\left[K_{j^*}\right] > 0$, where $\xrightarrow{D}$ denotes converging in distribution. It further leads to a conclusion that

$$\mathbb{P}\left(z_{t+N,j^*} - z_{t+N,j_{(r+1)}} > 0 \text{ for any } N > 0\right) > 0.$$

This means that $\mathbb{P}(j^* \in \mathcal{O}(\tau) \text{ for any } \tau \geq t) > 0$. Therefore, we have proved property 3.2.

S1.6: Proof of Property 4.1

In this section, we will prove Property 4.1 of the SRAS algorithm in Section 3.2.

Property 4.1: Assume $\lambda = 1$, $m_0 \to \infty$, all elements in the bandwidth vector $\boldsymbol{H}_j^{\mathcal{O}} \to \infty$ for any $\boldsymbol{H}_j^{\mathcal{O}} \in \mathcal{H}$ and all the data streams are exchangeable. When the process is IC, the distribution of the monitoring statistic Q_t will be free of the choice of $\Delta^{(t)}$.

Proof of Property 4.1:

According to equation (6) and the assumptions in the property, we derive that

$$Q_t = \frac{(2-\lambda)p}{\lambda\xi}\|\boldsymbol{v}_t\|^2 = \frac{p}{\xi}\|\boldsymbol{R}_t\|^2,$$

where $\xi = \frac{1}{m_0}\sum_{j=-m_0+1}^{0}\left\|\boldsymbol{R}_j\right\|^2$ and p are constant given historical IC data. Thus, the distribution of Q_t only depends on $\boldsymbol{R}_t$.

$$\boldsymbol{R}_t = \mathbb{E}_{\boldsymbol{X}}\left[U\left(\boldsymbol{M}_0\left(\widetilde{\boldsymbol{X}}_t - \boldsymbol{X}\right)\right)\right] = \mathbb{E}_{\boldsymbol{X}}[\|\boldsymbol{V}(t)\|^{-1}\boldsymbol{V}(t)],$$

where $\boldsymbol{V}(t) = \boldsymbol{M}_0\left(\widetilde{\boldsymbol{X}}_t - \boldsymbol{X}\right)$. Since exchangeable data streams are assumed, by definition, $\boldsymbol{M}_0$ should be a matrix with the same diagonal elements and same off-diagonal elements, respectively. Without loss of generality, assume any two different compensation coefficients $\Delta_1^{(t)}$ and $\Delta_2^{(t)}$ and thus the current observable layouts are assumed as $\mathcal{O}_1$ and $\mathcal{O}_2$, respectively. Similarly, denote the unobservable sets by $\mathcal{U}_1$ and $\mathcal{U}_2$ respectively. Therefore, we can derive the current augmented measurement vectors $\widetilde{\boldsymbol{X}}_t^{(1)}$ and $\widetilde{\boldsymbol{X}}_t^{(2)}$ respectively and reorder them such that

$$\widetilde{\boldsymbol{X}}_t^{(1)} = \begin{pmatrix} \boldsymbol{X}_t^{\mathcal{O}_1} \\ \boldsymbol{\mu}_0^{\mathcal{U}_1} \end{pmatrix},$$

$$\widetilde{\boldsymbol{X}}_t^{(2)} = \begin{pmatrix} \boldsymbol{X}_t^{\mathcal{O}_2} \\ \boldsymbol{\mu}_0^{\mathcal{U}_2} \end{pmatrix},$$

where $\boldsymbol{X}_t^{\mathcal{O}_1}$ and $\boldsymbol{X}_t^{\mathcal{O}_2}$ are the r-dimensional observed subsets of $\boldsymbol{X}_t$ with compensation coefficients $\Delta_1^{(t)}$ and $\Delta_2^{(t)}$, respectively. Similarly, $\boldsymbol{\mu}_0^{\mathcal{U}_1}$ and $\boldsymbol{\mu}_0^{\mathcal{U}_2}$ are the $(p-r)$-dimensional unobservable subsets of $\boldsymbol{\mu}_0$ with $\Delta_1^{(t)}$ and $\Delta_2^{(t)}$ respectively. Given that all the data streams are exchangeable, we obtain that $\boldsymbol{X}_t^{\mathcal{O}_1}$ has the same distribution as $\boldsymbol{X}_t^{\mathcal{O}_2}$ and $\boldsymbol{\mu}_0^{\mathcal{U}_1} = \boldsymbol{\mu}_0^{\mathcal{U}_2}$, and thus $\widetilde{\boldsymbol{X}}_t^{(1)}$ has the same distribution as $\widetilde{\boldsymbol{X}}_t^{(2)}$. Therefore, we conclude that with any compensation coefficient $\Delta^{(t)}$, the augmented measurement vector $\widetilde{\boldsymbol{X}}_t = \binom{\boldsymbol{X}_t^{\mathcal{O}}}{\boldsymbol{\mu}_0^{\mathcal{U}}}$ remains the same distribution. Further, the distribution of $\boldsymbol{V}(t) = \boldsymbol{M}_0(\widetilde{\boldsymbol{X}}_t - \boldsymbol{X})$ remains the same and so does $\boldsymbol{R}_t = \mathbb{E}_{\boldsymbol{X}}[\|\boldsymbol{V}(t)\|^{-1}\boldsymbol{V}(t)]$. Therefore, the distribution of the monitoring statistic $Q_t = \frac{p}{\xi}\|\boldsymbol{R}_t\|^2$ also remains the same. This way, we have proved this property that the distribution of the monitoring statistic Q_t is free of the choice of compensation coefficient $\Delta^{(t)}$.

S1.7: Proof of Property 4.2

In this section, we will prove Property 4.2 of the SRAS algorithm in Section 3.2.

Property 4.2: Assume $\lambda = 1$, $m_0 \to \infty$, all elements in the bandwidth vector satisfy $\boldsymbol{H}_j^{\mathcal{O}} \to \infty$ for any $\boldsymbol{H}_j^{\mathcal{O}} \in \mathcal{H}$, and all the data streams are exchangeable. Suppose we apply $\Delta^{(t)} = \Delta_0$, if $Q_t^{(1)} < K$ for the Aug(1) method and $\Delta^{(t)} = \begin{cases} \Delta_1, & \text{if } b_1 < Q_t^{(2)} < K, \\ \Delta_2, & \text{if } Q_t^{(2)} \le b_1, \end{cases}$ for the Aug(2) method with b_1 such that $b_1 = \left[Q_t^{(2)} \middle| \mathcal{O}^{(2)}(t) \subseteq \mathcal{I}(t)\right]_{q_1}$, where $\mathcal{I}(t)$ represents the set of IC data streams. Let the superscript "$^{(k)}$" represent relevant quantities for the Aug(k) method and the subscript "$_q$" represent relevant $100q$ percentile of the given variable. Suppose that after time t_0, there is a mean shift in only one data stream j^* such that $b_1 = \left[Q_t^{(2)} \middle| j^* \in \mathcal{O}^{(2)}(t)\right]_{q_1'}$ while the other data streams are IC, where $0 < q_1' < q_1 < 1$.

At any time $t > t_0$, the probability that the OC data stream j^* will be observed next time by the Aug(2) method is larger than that by the Aug(1) method, i.e., $\mathbb{P}\left(j^* \in \mathcal{O}^{(2)}(t+1)\right) > \mathbb{P}\left(j^* \in \mathcal{O}^{(1)}(t+1)\right)$ holds if: (i) the OC data stream j^* is unobserved, i.e., $j^* \in \mathcal{U}^{(k)}(t), k = 1,2$, given that $\Delta_2 > \Delta_1 = \Delta_0 > \mathbb{E}\left[\left(R_{t,i}^{(k)}\right)^2 \middle| i \in \mathcal{O}^{(k)}(t), \mathcal{O}^{(k)}(t) \subseteq \mathcal{I}(t)\right] - \mathbb{E}\left[\left(R_{t,j}^{(k)}\right)^2 \middle| j \in \mathcal{U}^{(k)}(t)\right], k = 1,2$; and (ii) the OC data stream j^* is observed, i.e., $j^* \in \mathcal{O}^{(k)}(t), k = 1,2$, given that $\Delta_1 < \Delta_2 = \Delta_0 < \mathbb{E}\left[\left(R_{t,j^*}^{(k)}\right)^2 \middle| j^* \in \mathcal{O}^{(k)}(t)\right] - \mathbb{E}\left[\left(R_{t,j}^{(k)}\right)^2 \middle| j \in \mathcal{U}^{(k)}(t)\right], k = 1,2.$

To prove Property 4.2, we consider the following two lemmas first:

Lemma 2.1: Assume $\lambda = 1$, $m_0 \to \infty$, all elements in the bandwidth vector $\boldsymbol{H}_j^{\mathcal{O}} \to \infty$ for any $\boldsymbol{H}_j^{\mathcal{O}} \in \mathcal{H}$ and all the data streams are exchangeable. Let the superscript "$^{(k)}$" represent relevant quantities for the Aug(k) method. Suppose all the observed data streams are IC in both methods, i.e., $\mathcal{O}^{(k)}(t) \subseteq \mathcal{I}(t), k = 1,2$, where $\mathcal{I}(t)$ represents the set of IC data streams at time t, and there exists one data stream j^* that is unobservable in both methods, i.e., $j^* \in \mathcal{U}^{(k)}(t), k = 1,2$ and $\mathcal{U}^{(1)}(t) \cap \mathcal{U}^{(2)}(t) \neq \emptyset$. Then $R_{t,j^*}^{(1)}$ and $R_{t,j^*}^{(2)}$ have the same distributions and so do $R_{t,j_{(r)}^{(1)}}^{(1)}$ and $R_{t,j_{(r)}^{(2)}}^{(2)}$.

Proof of Lemma 2.1:

Recall the augmented spatial rank at time t such that

$$\boldsymbol{R}_t = \mathbb{E}_{\boldsymbol{X}}\left[U\left(\boldsymbol{M}_0\left(\widetilde{\boldsymbol{X}}_t - \boldsymbol{X}\right)\right)\right] = \mathbb{E}_{\boldsymbol{X}}[\|\boldsymbol{V}(t)\|^{-1}\boldsymbol{V}(t)],$$

where $\boldsymbol{V}(t) = \boldsymbol{M}_0\left(\widetilde{\boldsymbol{X}}_t - \boldsymbol{X}\right)$. Since exchangeable data streams are assumed, by definition, $\boldsymbol{M}_0$ should be a matrix with the same diagonal elements and same off-diagonal elements, respectively. Recall that $j_{(1)}^{(k)}, \cdots, j_{(p)}^{(k)}, k = 1,2$ are the data stream indices such that $\tilde{L}_{t,j_{(1)}^{(k)}}^{(k)} \geq \tilde{L}_{t,j_{(2)}^{(k)}}^{(k)} \geq \cdots \geq \tilde{L}_{t,j_{(p)}^{(k)}}^{(k)}$ for the Aug$(k), k = 1,2$ methods, respectively. We can then derive the current augmented measurement vectors $\widetilde{\boldsymbol{X}}_t^{(1)}$ and $\widetilde{\boldsymbol{X}}_t^{(2)}$ respectively and reorder them according to indices $j_{(1)}^{(k)}, \cdots, j_{(p)}^{(k)}, k = 1,2$ such that

$$\widetilde{\boldsymbol{X}}_t^{(1)} = \begin{pmatrix} \boldsymbol{X}_t^{\mathcal{O}^{(1)}} \\ \boldsymbol{\mu}_0^{\mathcal{U}^{(1)}} \end{pmatrix},$$

$$\widetilde{\boldsymbol{X}}_t^{(2)} = \begin{pmatrix} \boldsymbol{X}_t^{\mathcal{O}^{(2)}} \\ \boldsymbol{\mu}_0^{\mathcal{U}^{(2)}} \end{pmatrix},$$

where $\boldsymbol{X}_t^{\mathcal{O}^{(1)}}$ and $\boldsymbol{X}_t^{\mathcal{O}^{(2)}}$ are the r-dimensional observed subsets of $\boldsymbol{X}_t$ for the Aug(1) and Aug(2) methods, respectively. Similarly, $\boldsymbol{\mu}_0^{\mathcal{U}^{(1)}}$ and $\boldsymbol{\mu}_0^{\mathcal{U}^{(2)}}$ are the $(p-r)$-dimensional unobservable subsets of $\boldsymbol{\mu}_0$ for the Aug(1) and Aug(2) methods, respectively. Given that all the data streams are exchangeable, $\boldsymbol{X}_t^{\mathcal{O}^{(1)}}$ should have the same distribution as $\boldsymbol{X}_t^{\mathcal{O}^{(2)}}$ and $\boldsymbol{\mu}_0^{\mathcal{U}^{(1)}} = \boldsymbol{\mu}_0^{\mathcal{U}^{(2)}}$, and thus $\widetilde{\boldsymbol{X}}_t^{(1)}$ should have the same distribution as $\widetilde{\boldsymbol{X}}_t^{(2)}$. Accordingly, the corresponding augmented spatial rank $\boldsymbol{R}_t^{(1)}$ should also have the same distribution as $\boldsymbol{R}_t^{(2)}$. Further, the corresponding $R_{t,j^*}^{(1)}$ and $R_{t,j^*}^{(2)}$ for $j^* \in \mathcal{U}^{(k)}(t), k = 1,2$ should thus have the same distributions and so do $R_{t,j_{(r)}^{(1)}}^{(1)}$ and $R_{t,j_{(r)}^{(2)}}^{(2)}$. Therefore, we have proved Lemma 2.1.

Lemma 2.2: Assume $\lambda = 1$, $m_0 \to \infty$, all elements in the bandwidth vector $\boldsymbol{H}_j^{\mathcal{O}} \to \infty$ for any $\boldsymbol{H}_j^{\mathcal{O}} \in \mathcal{H}$ and all the data streams are exchangeable when the process is IC. Suppose that after time t_0, there is a mean shift in one data stream j^* while the other data streams are IC. Let the superscript "$^{(k)}$" represent relevant quantities for the Aug(k) method. At any time $t > t_0$, assume that the OC data stream j^* is observed in both Aug(1) and Aug(2) methods, i.e., $j^* \in \mathcal{O}^{(k)}(t), k = 1,2$. Then $R_{t,j^*}^{(1)}$ and $R_{t,j^*}^{(2)}$ have the same distributions and so do $R_{t,j_{(r+1)}^{(1)}}^{(1)}$ and $R_{t,j_{(r+1)}^{(2)}}^{(2)}$.

Proof of Lemma 2.2:

Similar to Lemma 2.1, recall the augmented spatial rank at time t such that

$$\boldsymbol{R}_t = \mathbb{E}_{\boldsymbol{X}}\left[U\left(\boldsymbol{M}_0\left(\widetilde{\boldsymbol{X}}_t - \boldsymbol{X}\right)\right)\right] = \mathbb{E}_{\boldsymbol{X}}[\|\boldsymbol{V}(t)\|^{-1}\boldsymbol{V}(t)],$$

where $\boldsymbol{V}(t) = \boldsymbol{M}_0\left(\widetilde{\boldsymbol{X}}_t - \boldsymbol{X}\right)$. Since exchangeable data streams are assumed, by definition, $\boldsymbol{M}_0$ should be a matrix with the same diagonal elements and same off-diagonal elements, respectively. Recall that $j_{(1)}^{(k)}, \cdots, j_{(p)}^{(k)}, k = 1,2$ are the data stream indices such that $\tilde{L}_{t,j_{(1)}^{(k)}}^{(k)} \geq \tilde{L}_{t,j_{(2)}^{(k)}}^{(k)} \geq \cdots \geq \tilde{L}_{t,j_{(p)}^{(k)}}^{(k)}$ for Aug$(k), k = 1,2$ methods, respectively. At any time $t > t_0$, we can then derive the current augmented measurement vectors $\widetilde{\boldsymbol{X}}_t^{(1)}$ and $\widetilde{\boldsymbol{X}}_t^{(2)}$ respectively and reorder the observed OC data stream j^* in the first location and the remaining ones according to indices $j_{(1)}^{(k)}, \cdots, j_{(p)}^{(k)}, k = 1,2$ such that

$$\widetilde{\boldsymbol{X}}_t^{(1)} = \begin{pmatrix} \boldsymbol{X}_t^{\mathcal{O}^{(1)}} \\ \boldsymbol{\mu}_0^{\mathcal{U}^{(1)}} \end{pmatrix},$$

$$\widetilde{\boldsymbol{X}}_t^{(2)} = \begin{pmatrix} \boldsymbol{X}_t^{\mathcal{O}^{(2)}} \\ \boldsymbol{\mu}_0^{\mathcal{U}^{(2)}} \end{pmatrix},$$

where $\boldsymbol{X}_t^{\mathcal{O}^{(1)}}$ and $\boldsymbol{X}_t^{\mathcal{O}^{(2)}}$ are the r-dimensional observed subsets of $\boldsymbol{X}_t$ for the Aug(1) and Aug(2) methods, respectively. Similarly, $\boldsymbol{\mu}_0^{\mathcal{U}^{(1)}}$ and $\boldsymbol{\mu}_0^{\mathcal{U}^{(2)}}$ are the $(p-r)$-dimensional unobservable subsets of $\boldsymbol{\mu}_0$ for the Aug(1) and Aug(2) methods, respectively. Given that all the data streams are exchangeable when the process is IC and there is only one OC data stream $j^* \in \mathcal{O}^{(k)}(t), k = 1,2$ at time $t > t_0$, $\boldsymbol{X}_t^{\mathcal{O}^{(1)}}$ should have the same distribution as $\boldsymbol{X}_t^{\mathcal{O}^{(2)}}$ and $\boldsymbol{\mu}_0^{\mathcal{U}^{(1)}} = \boldsymbol{\mu}_0^{\mathcal{U}^{(2)}}$, and thus $\widetilde{\boldsymbol{X}}_t^{(1)}$ should have the same distribution as $\widetilde{\boldsymbol{X}}_t^{(2)}$. Accordingly, the corresponding augmented spatial rank $\boldsymbol{R}_t^{(1)}$ should also have the same distribution as $\boldsymbol{R}_t^{(2)}$. Further, the corresponding $R_{t,j^*}^{(1)}$ and $R_{t,j^*}^{(2)}$ as well as $R_{t,j_{(r)}^{(1)}}^{(1)}$ and $R_{t,j_{(r)}^{(2)}}^{(2)}$ should thus have the same distributions, respectively. Therefore, we have proved Lemma 2.2.

Proof of Property 4.2:

First, for the statement (i), consider that at time $t > t_0$, the OC data stream X_{t,j^*} is unobservable (i.e. $j^* \in \mathcal{U}^{(k)}(t), k = 1,2$), and $\Delta_2 > \Delta_1 = \Delta_0 > \mathbb{E}\left[\left(R_{t,i}^{(k)}\right)^2 \middle| i \in \mathcal{O}^{(k)}(t), \mathcal{O}^{(k)}(t) \subseteq \mathcal{I}(t)\right] - \mathbb{E}\left[\left(R_{t,j}^{(k)}\right)^2 \middle| j \in \mathcal{U}^{(k)}(t)\right], k = 1,2.$

As $\lambda = 1$, we have that $\boldsymbol{z}_t = \tilde{\boldsymbol{L}}_t$. Let $\tilde{\boldsymbol{L}}_t^{(k)}, k = 1,2$ denote the squared spatial rank statistics at time t with Aug(k) method respectively. Then by definition, $\mathcal{O}^{(k)}(t) \subseteq \mathcal{I}(t)$ and $\tilde{L}_{t,j^*}^{(k)} < \tilde{L}_{t,i}^{(k)}, \forall i \in \mathcal{O}^{(k)}(t), k = 1,2$ respectively. Recall that $j_{(1)}^{(k)}, \cdots, j_{(p)}^{(k)}, k = 1,2$ are the data stream indices such that $\tilde{L}_{t,j_{(1)}^{(k)}}^{(k)} \geq \tilde{L}_{t,j_{(2)}^{(k)}}^{(k)} \geq \cdots \geq \tilde{L}_{t,j_{(p)}^{(k)}}^{(k)}$ with Aug$(k), k = 1,2$ method. Then the difference between the augmented squared spatial rank statistics on data stream j^* and $j_{(r)}^{(1)}$ with Aug(1) method at time t is defined as

$$\tilde{L}_{t,j^*}^{(1)} - \tilde{L}_{t,j_{(r)}^{(1)}}^{(1)} = \left(R_{t,j^*}^{(1)}\right)^2 + \Delta_0 - \left(R_{t,j_{(r)}^{(1)}}^{(1)}\right)^2.$$

Similarly, the difference between the augmented squared spatial rank statistics on data stream j^* and $j_{(r)}^{(2)}$ with Aug(2) method at time t is defined as

$$\tilde{L}_{t,j^*}^{(2)} - \tilde{L}_{t,j_{(r)}^{(2)}}^{(2)} = \left(R_{t,j^*}^{(2)}\right)^2 + \Delta^{(t)} - \left(R_{t,j_{(r)}^{(2)}}^{(2)}\right)^2.$$

Note that their means satisfy that $\mathbb{E}\left[\tilde{L}_{t,j^*}^{(2)} - \tilde{L}_{t,j_{(r)}^{(2)}}^{(2)}\right] > \mathbb{E}\left[\tilde{L}_{t,j^*}^{(1)} - \tilde{L}_{t,j_{(r)}^{(1)}}^{(1)}\right] > 0$. Therefore, the probability that the difference with the Aug(k) method is positive should be larger than zero such that $\mathbb{P}\left(\tilde{L}_{t,j^*}^{(k)} - \tilde{L}_{t,j_{(r)}^{(1)}}^{(k)} > 0\right) > 0, k = 1,2$. Based on $\Delta_2 > \Delta_1 = \Delta_0$ and Lemma 2.1 that $R_{t,j^*}^{(1)}$ and $R_{t,j^*}^{(2)}$ have the same distributions and so do $R_{t,j_{(r)}^{(1)}}^{(1)}$ and $R_{t,j_{(r)}^{(2)}}^{(2)}$, it leads to the conclusion that

$$\mathbb{P}\left(\tilde{L}_{t,j^*}^{(2)} - \tilde{L}_{t,j_{(r)}^{(2)}}^{(2)} > 0\right) > \mathbb{P}\left(\tilde{L}_{t,j^*}^{(1)} - \tilde{L}_{t,j_{(r)}^{(1)}}^{(1)} > 0\right).$$

Therefore, we have proved the statement (i) such that $\mathbb{P}\left(j^* \in \mathcal{O}^{(2)}(t+1)\right) > \mathbb{P}\left(j^* \in \mathcal{O}^{(1)}(t+1)\right)$.

Next, similarly for the statement (ii), consider that at time $t > t_0$, the OC data stream X_{t,j^*} is observable (i.e. $j^* \in \mathcal{O}^{(k)}(t)$), and $\Delta_1 < \Delta_2 = \Delta_0 < \mathbb{E}\left[\left(R_{t,j^*}^{(k)}\right)^2 \middle| j^* \in \mathcal{O}^{(k)}(t)\right] - \mathbb{E}\left[\left(R_{t,j}^{(k)}\right)^2 \middle| j \in \mathcal{U}^{(k)}(t)\right], k = 1,2$. Recall that $j_{(1)}^{(k)}, \cdots, j_{(p)}^{(k)}, k = 1,2$ are the data stream indices such that $\tilde{L}_{t,j_{(1)}^{(k)}}^{(k)} \geq \tilde{L}_{t,j_{(2)}^{(k)}}^{(k)} \geq \cdots \geq \tilde{L}_{t,j_{(p)}^{(k)}}^{(k)}$ with Aug$(k), k = 1,2$ method. Then the difference between the augmented squared spatial rank statistics on data stream j^* and $j_{(r+1)}^{(1)}$ with Aug(1) method at time t is defined as

$$\tilde{L}_{t,j^*}^{(1)} - \tilde{L}_{t,j_{(r+1)}^{(1)}}^{(1)} = \left(R_{t,j^*}^{(1)}\right)^2 - \left(R_{t,j_{(r+1)}^{(1)}}^{(1)}\right)^2 - \Delta_0.$$

Similarly, the difference between the augmented squared spatial rank statistics on data stream j^* and $j_{(r+1)}^{(2)}$ with Aug(2) method at time t is defined as

$$\tilde{L}_{t,j^*}^{(2)} - \tilde{L}_{t,j_{(r+1)}^{(2)}}^{(2)} = \left(R_{t,j^*}^{(2)}\right)^2 - \left(R_{t,j_{(r+1)}^{(2)}}^{(2)}\right)^2 - \Delta^{(t)}.$$

Note that their means satisfy that $\mathbb{E}\left[\tilde{L}_{t,j^*}^{(2)} - \tilde{L}_{t,j_{(r+1)}^{(2)}}^{(2)}\right] > \mathbb{E}\left[\tilde{L}_{t,j^*}^{(1)} - \tilde{L}_{t,j_{(r+1)}^{(1)}}^{(1)}\right] > 0$. Therefore, the probability that the difference with the Aug(k) method is positive should be larger than zero such that $\mathbb{P}\left(\tilde{L}_{t,j^*}^{(k)} - \tilde{L}_{t,j_{(r+1)}^{(1)}}^{(k)} > 0\right) > 0, k = 1,2$. Based on $\Delta_1 < \Delta_2 = \Delta_0$ and Lemma 2.2 that $R_{t,j^*}^{(1)}$ and $R_{t,j^*}^{(2)}$ have the same distributions and so do $R_{t,j_{(r+1)}^{(1)}}^{(1)}$ and $R_{t,j_{(r+1)}^{(2)}}^{(2)}$, it leads to the conclusion that

$$\mathbb{P}\left(\tilde{L}_{t,j^*}^{(2)} - \tilde{L}_{t,j_{(r+1)}^{(2)}}^{(2)} > 0\right) > \mathbb{P}\left(\tilde{L}_{t,j^*}^{(1)} - \tilde{L}_{t,j_{(r+1)}^{(1)}}^{(1)} > 0\right).$$

Therefore, we have proved the statement (ii) such that $\mathbb{P}\left(j^* \in \mathcal{O}^{(2)}(t+1)\right) > \mathbb{P}\left(j^* \in \mathcal{O}^{(1)}(t+1)\right)$. To this end, we have completed the proof of the property 4.2.

Corollary 2: Assume $\lambda = 1$, $m_0 \to \infty$, all elements in the bandwidth vector satisfy $\boldsymbol{H}_j^{\mathcal{O}} \to \infty$ for any $\boldsymbol{H}_j^{\mathcal{O}} \in \mathcal{H}$, and all the data streams are exchangeable. Suppose for the Aug(k) method with $k > 1$, we apply $\Delta^{(t,k)} = \begin{cases} \Delta_1^{(k)}, \text{ if } b_1^{(k)} < Q_t^{(k)} < K, \\ \Delta_2^{(k)}, \text{ if } b_2^{(k)} < Q_t^{(k)} \le b_1^{(k)}, \\ \quad \vdots \\ \Delta_k^{(k)}, \text{ if } 0 \le Q_t^{(k)} \le b_{k-1}^{(k)}, \end{cases}$ with $b_l^{(k)} = \left[Q_t^{(k)} \middle| \mathcal{O}^{(k)}(t) \subseteq \mathcal{I}(t)\right]_{q_l^{(k)}}$, for $l = 1, \cdots, k-1$ with $0 < q_{k-1}^{(k)} < \cdots < q_1^{(k)} < 1$, where $\mathcal{I}(t)$ represents the set of IC data streams. Let the superscript "$^{(k)}$" represent relevant quantities for the Aug(k) method and the subscript "$_q$" represent relevant the $100q$ percentile of a given variable. Suppose that after time t_0, there is a mean shift in only one data stream j^* such that $b_l^{(k)} = \left[Q_t^{(k)} \middle| j^* \in \mathcal{O}^{(k)}(t)\right]_{q_l^{(k)\prime}}$, where $0 < q_l^{(k)\prime} < q_l^{(k)} < 1$, for $l = 1, \cdots, k-1$. At any time $t > t_0$, the probability that the OC data stream j^* will be observed next time by the Aug(k_0+1) method is larger than that by the Aug(k_0) method, i.e., $\mathbb{P}\left(j^* \in \mathcal{O}^{(k_0+1)}(t+1)\right) > \mathbb{P}\left(j^* \in \mathcal{O}^{(k_0)}(t+1)\right)$ holds if: (i) the OC data stream j^* is unobserved, i.e., $j^* \in \mathcal{U}^{(k_0)}(t)$ and $j^* \in \mathcal{U}^{(k_0+1)}(t)$, given that $\Delta_{k_0+1} > \Delta_{k_0} \ge \cdots \ge \Delta_1 > \mathbb{E}\left[\left(R_{t,i}^{(k)}\right)^2 \middle| i \in \mathcal{O}^{(k)}(t), \mathcal{O}^{(k)}(t) \subseteq \mathcal{I}(t)\right] - \mathbb{E}\left[\left(R_{t,j}^{(k)}\right)^2 \middle| j \in \mathcal{U}^{(k)}(t)\right], k = k_0, k_0+1$, where $\Delta_l = \Delta_l^{(k_0)} = \Delta_l^{(k_0+1)}$, for $l = 1, \cdots, k_0$, $\Delta_{k_0+1} = \Delta_{k_0+1}^{(k_0+1)}$ and $q_l^{(k_0)} = q_l^{(k_0+1)}$, for $l = 1, \cdots, k_0 - 1$; and (ii) the OC data stream j^* is observed, i.e., $j^* \in$

$\mathcal{O}^{(k_0)}(t)$ and $j^* \in \mathcal{O}^{(k_0+1)}(t)$ given that $\Delta_1 < \Delta_2 \leq \cdots \leq \Delta_{k_0+1} < \mathbb{E}\left[\left(R_{t,j^*}^{(k)}\right)^2 \middle| j^* \in \mathcal{O}^{(k)}(t)\right] - \mathbb{E}\left[\left(R_{t,j}^{(k)}\right)^2 \middle| j \in \mathcal{U}^{(k)}(t)\right], k = k_0, k_0+1$, where $\Delta_{l+1} = \Delta_l^{(k_0)} = \Delta_{l+1}^{(k_0+1)}$, for $l = 1, \cdots, k_0$, $\Delta_1 = \Delta_1^{(k_0+1)}$ and $q_{l+1}^{(k_0)} = q_l^{(k_0+1)}$, for $l = 1, \cdots, k_0 - 1$.

Proof of Corollary 2:

The proof of this corollary is very similar to Property 4.2 by replacing Aug(1) and Aug(2) by Aug(k_0) and Aug(k_0+1), respectively. First, the results shown in the Lemmas 2.1 and 2.2 can be extended to the Aug(k_0) and Aug(k_0+1) methods.

Next, for the statement (i) when the OC data stream X_{t,j^*} is unobservable at time $t > t_0$, given that $\Delta_{k_0+1} > \Delta_{k_0} \geq \cdots \geq \Delta_1$ where $\Delta_l = \Delta_l^{(k_0)} = \Delta_l^{(k_0+1)}$, for $l = 1, \cdots, k_0$, $\Delta_{k_0+1} = \Delta_{k_0+1}^{(k_0+1)}$ and $q_l^{(k_0)} = q_l^{(k_0+1)}$, for $l = 1, \cdots, k_0 - 1$, we can find that $\tilde{L}_{t,j^*}^{(k_0+1)} - \tilde{L}_{t,j_{(r)}^{(k_0+1)}}^{(k_0+1)} > \tilde{L}_{t,j^*}^{(k_0)} - \tilde{L}_{t,j_{(r)}^{(k_0)}}^{(k_0)}$, which leads to the conclusion that $\mathbb{P}\left(\tilde{L}_{t,j^*}^{(k_0+1)} - \tilde{L}_{t,j_{(r)}^{(k_0+1)}}^{(k_0+1)} > 0\right) > \mathbb{P}\left(\tilde{L}_{t,j^*}^{(k_0)} - \tilde{L}_{t,j_{(r)}^{(k_0)}}^{(k_0)} > 0\right)$. For the statement (ii) when the OC data stream X_{t,j^*} is observable at time $t > t_0$, given that $\Delta_1 < \Delta_2 \leq \cdots \leq \Delta_{k_0+1}$ where $\Delta_{l+1} = \Delta_l^{(k_0)} = \Delta_{l+1}^{(k_0+1)}$, for $l = 1, \cdots, k_0$, $\Delta_1 = \Delta_1^{(k_0+1)}$ and $q_{l+1}^{(k_0)} = q_l^{(k_0+1)}$, for $l = 1, \cdots, k_0 - 1$, we can find that $\tilde{L}_{t,j^*}^{(k_0+1)} - \tilde{L}_{t,j_{(r+1)}^{(k_0+1)}}^{(k_0+1)} > \tilde{L}_{t,j^*}^{(k_0)} - \tilde{L}_{t,j_{(r+1)}^{(k_0)}}^{(k_0)}$, which leads to the conclusion that $\mathbb{P}\left(\tilde{L}_{t,j^*}^{(k_0+1)} - \tilde{L}_{t,j_{(r+1)}^{(k_0+1)}}^{(k_0+1)} > 0\right) > \mathbb{P}\left(\tilde{L}_{t,j^*}^{(k_0)} - \tilde{L}_{t,j_{(r+1)}^{(k_0)}}^{(k_0)} > 0\right)$. Combining the two conclusions, we have shown that $\mathbb{P}\left(j^* \in \mathcal{O}^{(k_0+1)}(t+1)\right) > \mathbb{P}\left(j^* \in \mathcal{O}^{(k_0)}(t+1)\right)$ in two scenarios and thus completed the proof of the Corollary 2.

S2: Parameter settings

In this section, we will show the selection of the parameters $\mathcal{H}$, λ, $\mathcal{M}$ involved in the proposed SRAS algorithm in detail.

(i) The selection of $\mathcal{H}$: Let $\mathcal{S}$ denote the set of all candidate sampling layouts and thus $\mathcal{O}(t) \in \mathcal{S}$ for any t and $|\mathcal{S}| = \binom{p}{r}$. $\mathcal{H}$ denotes the set of candidate bandwidth vectors such that $\mathcal{H} = \{\boldsymbol{H}_j^{\mathcal{O}} : j \in \mathcal{P}, \mathcal{O} \in$

$\mathcal{S}\}$, and thus $\boldsymbol{H}_j^{\mathcal{O}(t)} \in \mathcal{H}$ for any t. As widely studied in literature, the bandwidth determines the smoothness of estimator, i.e., a large value leads to a smoother estimation with smaller variance but higher bias while a smaller value results in a more erratic estimation with higher variance but lower bias, and thus the value of the bandwidth can be neither too small nor too large. To determine the bandwidth vectors in $\mathcal{H}$ offline based on the historical observations, we apply the conventional leave-one-out cross-validation (CV) method to handle the bias-variance tradeoff (Schindler 2012) by solving a minimization problem such that $\min CV\left(\boldsymbol{H}_j^{\mathcal{O}}\right)$, where $\boldsymbol{H}_j^{\mathcal{O}}$ is the bandwidth vector to be determined and $CV(\cdot)$ is the objective function based on a candidate CV criterion. Since solving the minimization problem for all $\binom{p}{r}$ candidate $\boldsymbol{H}_j^{\mathcal{O}} \in \mathcal{H}$ is a very expensive job, we suggest instead solving for a p-dimensional vector $\boldsymbol{H} = \left(h_1, \cdots, h_p\right)'$ and then construct $\boldsymbol{H}_j^{\mathcal{O}}$ by selecting the r-dimensional observed subset of $\boldsymbol{H}$, i.e., $\boldsymbol{H}_j^{\mathcal{O}} = \{h_k, k \in \mathcal{O}\}, \forall j \in \mathcal{P}$, which shows robust practical performance and significantly reduces the complexity to the estimation of a single bandwidth vector $\boldsymbol{H}$.

In the proposed method, we extend the Jackknife nonparametric residuals to multivariate case and define the objective function $CV(\cdot)$ based on the Mean-Squared Forecast Error (MSFE) criterion (Köhler, Schindler, and Sperlich 2014) as

$$CV(\boldsymbol{H}) = \widehat{MSFE}(\boldsymbol{H}) = \frac{1}{m_0} \sum_{i=-m_0+1}^{0} \sum_{d \in \mathcal{P}} \tilde{e}_{i,d}^2(\boldsymbol{H}),$$

where $\tilde{e}_{i,d}(\boldsymbol{H})$ is the Jackknife nonparametric residual for $X_{i,d}, d \in \mathcal{P}, i \in \mathcal{T} := \{-m_0 + 1, -m_0 + 2, \cdots, 0\}$ based on all the m_0 IC historical observations $\boldsymbol{X}_i$ such that

$$\tilde{e}_{i,d}(\boldsymbol{H}) = X_{i,d} - \hat{X}_{i,d}, \qquad i \in \mathcal{T},$$

where $\hat{X}_{i,d}$ is the leave-one-out NW estimation such that

$$\hat{X}_{i,d} = \frac{\sum_{t \in \mathcal{T} \setminus \{i\}} K_{\boldsymbol{H}_{-d}}\left(\boldsymbol{X}_{i,-d} - \boldsymbol{X}_{t,-d}\right) \times X_{t,d}}{\sum_{t \in \mathcal{T} \setminus \{i\}} K_{\boldsymbol{H}_{-d}}\left(\boldsymbol{X}_{i,-d} - \boldsymbol{X}_{t,-d}\right)}, \qquad i \in \mathcal{T}.$$

Here, $\boldsymbol{H}_{-d} = \{h_k, k \in \mathcal{P} \setminus \{d\}\}$ and $\boldsymbol{X}_{i,-d} = \{X_{ik}, k \in \mathcal{P} \setminus \{d\}\}, i \in \mathcal{T}$. Consequently, we can derive the $CV(\cdot)$ based on MSFE criterion, and thus the best bandwidth vector $\widehat{\boldsymbol{H}}$ is obtained from $\widehat{\boldsymbol{H}} = \arg\min_{\boldsymbol{H}} CV(\boldsymbol{H})$.

(ii) The selection of λ: λ is the EWMA smoothing weight, which has been widely studied in the literature such that a smaller λ leads to a quicker detection of smaller shifts and a larger λ accelerates the detection of larger shifts. In practice, we suggest choosing $\lambda \in [0.05,0.2]$ in general and a smaller $\lambda \in [0.025,0.1]$ if there is evidence that the underlying distribution is very skewed or the number of data streams is very large (Zou et al. 2012).

(iii) The selection of $\mathcal{M} = \{k, b_1, \cdots, b_{k-1}, \Delta_1, \cdots, \Delta_k\}$: $\mathcal{M}$ is the set of parameters of the piecewise compensation coefficient function $\Delta(\cdot)$ in equation (11). Intuitively, a larger k will lead to a more intelligent balance between the connection of the monitoring and sampling schemes, whereas the parameter selection effort and computation cost could also be increased. Yet, it should be noted that the value selection of Δ_j has a significant effect on the sampling performance and the sophisticated parameters could be selected based on a simpler selection applying $\mathrm{Aug}(1)$ method when $k = 1$. Properties 3.1-4.2 in Section 3.2 shed light on the selection of Δ_j. In practice, as observed in Section 4.1, the $\mathrm{Aug}(2)$ method (i.e., $k = 2$) already enables a comprehensively better performance and thus is recommended, especially when the number of the observable data streams are extremely small. The exact values of Δ_j can be selected based on simulation results with $k = 1$, the discussion of which can be found in Section 4.1. Accordingly, as discussed in Section 3.1.4 that the compensation coefficient $\Delta(\cdot)$ is a non-increasing function of the monitoring statistic Q_t, the thresholds b_j can then be set based on the percentiles of Q_t. Generally, 50-90 percentiles are recommended for b_1 in the $\mathrm{Aug}(2)$ method, which is further discussed in Section 4.1.

S3: Simulation study regarding estimation performance under different OC scenarios

In this section, we will conduct further simulation on different choices of bandwidth parameter under different types of shifts to justify the estimation performance of the proposed method. For the augmentation of unobserved data streams, we generally adopt the NW estimation with the bandwidth selected through the procedure as discussed in Section 3.3. In this study, we also consider augmenting the unobserved data streams with their IC sample mean, i.e., the situation where the estimation converges to the IC sample mean as the bandwidth goes to infinity (shown in Property 1.1 in Section

3.2). The estimation performance with different choices of bandwidth parameter is compared under the SPC setting, and ARL_1 is adopted as metric to validate the performance in this study. Without loss of generality, we assume the data streams follow exchangeable normal distribution as described in Section 4.2.1 and there are $r = 4, 6$ and 8 observable data streams respectively. Specifically, we set $h = 1.1$ for NW estimation and parameters $\Delta = 0.1, \lambda = 0.1$. One randomly selected data stream has an OC shift of magnitude δ and different types of shifts are considered: (i) abrupt shift, i.e., a mean shift δ occurring abruptly at time $\tau + 1$ based on equation (1); and (ii) smooth shift where a mean shift δ occurring slowly and smoothly from time $\tau + 1$ to $\tau + s$ and remaining unchanged afterwards, i.e., $\delta(t) = \begin{cases} \frac{t-\tau}{s}\delta, t = \tau + 1, \cdots, \tau + s \\ \delta, \ \ t = \tau + s + 1, \cdots \end{cases}$, where $\delta(t)$ denotes the mean shift at time $t > \tau$. Different values of s imply different slopes of shift, in that larger s implies slower and smoother shift. The abrupt mean shift can be considered as a special case of smooth shift when $s = 1$. For each estimation method, three magnitudes of mean shift $\delta = 1,2$ and 3 are considered and the monitoring performance under four different values of $s = 1,5,10,20$ are evaluated for each δ. Table 8 summarizes the simulation results.

Table 8. ARL_1 and the standard errors in parentheses of the SRAS algorithm with exchangeable multivariate normal distribution under different numbers of observable data streams r, mean shifts δ and values of s for different estimation methods.

		NW estimation				**IC sample mean**			
		$s = 1$	$s = 5$	$s = 10$	$s = 20$	$s = 1$	$s = 5$	$s = 10$	$s = 20$
$r = 4$	$\delta = 1$	16.64 (0.15)	18.96 (0.15)	22.16 (0.15)	28.62 (0.17)	19.36 (0.18)	21.91 (0.19)	24.82 (0.19)	31.01 (0.20)
	$\delta = 2$	6.28 (0.03)	8.67 (0.03)	11.43 (0.04)	16.54 (0.05)	7.07 (0.03)	9.64 (0.04)	12.31 (0.05)	17.54 (0.06)
	$\delta = 3$	4.91 (0.01)	6.83 (0.02)	9.06 (0.03)	13.10 (0.04)	5.33 (0.02)	7.49 (0.03)	9.77 (0.03)	13.91 (0.04)
$r = 6$	$\delta = 1$	13.79 (0.09)	15.90 (0.09)	18.64 (0.10)	24.73 (0.11)	15.21 (0.11)	17.45 (0.12)	20.41 (0.11)	26.30 (0.12)
	$\delta = 2$	5.98 (0.02)	8.00 (0.02)	10.50 (0.03)	15.32 (0.04)	6.43 (0.02)	8.46 (0.02)	11.02 (0.03)	15.93 (0.04)
	$\delta = 3$	4.58 (0.01)	6.30 (0.01)	8.48 (0.02)	12.16 (0.03)	4.86 (0.01)	6.66 (0.02)	8.88 (0.02)	12.70 (0.03)
$r = 8$	$\delta = 1$	13.17 (0.07)	15.07 (0.08)	17.75 (0.08)	23.19 (0.09)	13.83 (0.08)	15.85 (0.08)	18.36 (0.08)	24.20 (0.09)
	$\delta = 2$	6.14 (0.01)	7.94 (0.02)	10.35 (0.02)	15.00 (0.04)	6.39 (0.02)	8.20 (0.02)	10.63 (0.02)	15.26 (0.04)
	$\delta = 3$	4.66 (0.01)	6.29 (0.01)	8.40 (0.02)	11.96 (0.03)	4.86 (0.01)	6.44 (0.01)	8.59 (0.02)	12.20 (0.03)

From Table 8, it can be easily seen that the utilization of the NW estimation outperforms using IC sample mean in every scenario, especially when the number of observed data streams r and the

magnitude of mean shift δ are relatively small. This is expected since the NW estimation better augments the unobserved data stream utilizing interrelationship information among data streams based on not only the IC historical data but also the currently observed data streams, thus leading to more powerful detection capability. Besides, comparing different shift scenarios with different ss, it is not surprising that large s leads to increased detection delay as the shift occurs smoothly. However, the differences between NW estimation and IC sample mean with respect to the value of s are similar, indicating that different types of shifts do not affect the estimation performance.

Additionally, the computational time under different scenarios assuming there are $r = 6$ observable data streams is also tabulated in Table 9. From the results, it can be seen that the computational time is similar under different values of s, also indicating that different types of shifts do not affect the estimation performance. The computational time through using the IC sample mean is reduced compared to that with the utilization of the NW estimation in every scenario, which is expected since the NW estimation requires more computation due to relatively sophisticated data augmentation through interrelationship analysis. To this end, based on the results in Table 8 and 9, the bandwidth value can be selected to be very large to ease the computation cost (e.g., the augmentation converges to the simple IC sample mean as the bandwidth goes to infinity), whose computational time is significantly reduced but the monitoring performance is still acceptable though with slight scarification.

Table 9. Computational time (milliseconds) with the standard errors in parentheses of the SRAS algorithm with exchangeable multivariate normal distribution under different mean shifts δ and values of s for different estimation methods.

		NW estimation				**IC sample mean**			
		$s = 1$	$s = 5$	$s = 10$	$s = 20$	$s = 1$	$s = 5$	$s = 10$	$s = 20$
$r = 6$	$\delta = 1$	0.09 (0.09)	0.09 (0.09)	0.09 (0.07)	0.09 (0.07)	0.03 (0.01)	0.02 (0.01)	0.02 (0.01)	0.02 (0.01)
	$\delta = 2$	0.09 (0.08)	0.09 (0.07)	0.09 (0.08)	0.09 (0.07)	0.02 (0.01)	0.02 (0.01)	0.02 (0.01)	0.02 (0.01)
	$\delta = 3$	0.09 (0.08)	0.09 (0.07)	0.09 (0.07)	0.09 (0.07)	0.02 (0.01)	0.02 (0.01)	0.02 (0.01)	0.02 (0.01)

S4: Case study of COVID-19 pandemic surveillance

In this section, we conduct another case study of online pandemic surveillance in IoT enabled healthcare systems to evaluate the performance of the proposed SRAS algorithm. Such systems collect patient data and transmit them to a fusion center for real-time analysis (Otoom et al. 2020), which assists with

continuous pandemic surveillance and quick response to the outbreak such as the subsequent policy implementation of the government (Darshan and Anandakumar 2015; Ukil et al. 2016). Although pandemic outbreak does not occur frequently, considerable amount of data from a huge number of individuals need to be transmitted to and processed at the fusion center, which is not cost-effective and places a heavy stress on the processing capability of the system. To enable frequent assessment of the pandemic and reduce processing costs, we instead adaptively determine the subset of data to be processed at each acquisition time for surveillance. Moreover, it should be noted that the collected data related to pandemic risk usually follow heterogenous distributions and may involve complex correlations among incidences due to transmission (Otoom et al. 2020), which requires nonparametric methods for robust and satisfactory performance. In this case study, we adopt a dataset of COVID-19 testing across the Florida from February to June, 2020. Each testing record includes information about the corresponding county, zip code, testing result (positive or negative), and the date of the test result, etc. In this study, we regard the daily COVID-19 positive rate at a zip code region as a data stream. After removing the regions with less than 100 testing cases, we retain 54 zip codes in the Broward county across 114 days from March 2 to June 23, 2020. Figure 4 visualizes the selected COVID-19 positive rates for the 54 data streams over 114 time frames. We can see that there is an abrupt increase of the data streams starting from time $t = 88$ approximatly, which can be regarded as the change point.

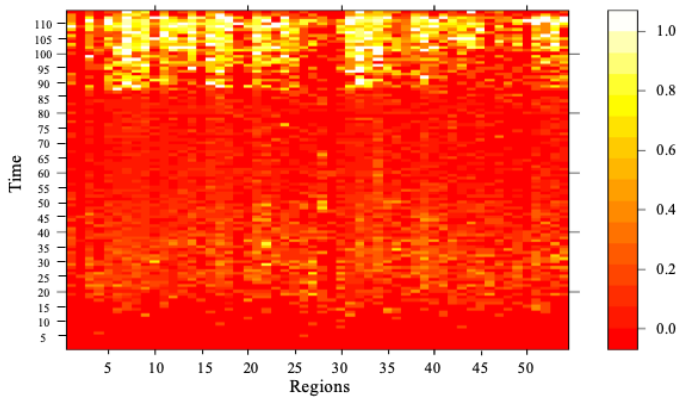


Figure 4. COVID-19 positive rates across different regions over time

Without loss of generality, the desired IC ARL is set to 200 and the control limits are obtained via the bootstrap procedure based on the dataset from $t = 1{\sim}30$. We assume that there are only $r = 10$ (18.52%) data streams that can be observed at each time, and the dataset from $t = 31{\sim}79$ is selected as the historical IC data. To demonstrate the performance of the SRAS algorithm, we compare

it with the NAS and the FULL algorithms and the parameters are selected as follows: (i) $h = 3, \Delta = 0.01, \lambda = 0.1$ for the SRAS algorithm; (ii) $k = 0.2, \Delta = \frac{1}{200}$ for the NAS algorithm; (iii) $\lambda = 0.1$ for the FULL method. Figure 5 illustrates the monitoring statistics of the three algorithms over time $t = 80{\sim}114$ with the control limits $K = 560,\ 2584$ and 120, respectively. As expected, the FULL method has the quickest response to the changes among the data streams at $t = 91$ since it assumes that full observations are available. Although only partial data streams are observable, the SRAS algorithm is still able to detect the abrupt change quickly at $t = 98$. After triggering an alarm, the monitoring statistic keeps increasing and remains above the control limit, indicating that the process is indeed OC. The NAS algorithm triggers an OC alarm at $t = 104$ afterwards. This is probably because the anti-rank based statistics is highly distorted given the heterogeneous distributions of the data streams, thus leading to deteriorated monitoring performance. Considering that only $r = 10$ data streams are observable, the SRAS algorithm shows a satisfactory performance for the pandemic surveillance. Table 10 further summarizes the computational time of the three algorithms, and the results are similar to those in Section 5. In particular, the NAS algorithm runs the fastest while the computational time of the SRAS algorithm is comparable to the FULL method.

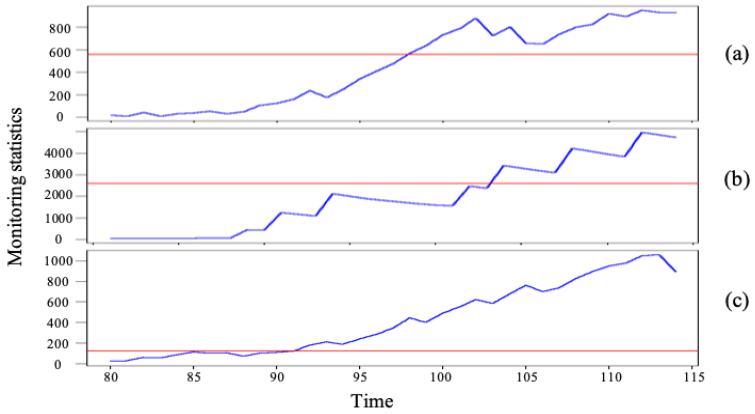


Figure 5. Monitoring statistics over time of different algorithms for the pandemic surveillance: (a) the SRAS algorithm, (b) the NAS algorithm, and (c) the FULL method.

Table 10. Comparison of computational time (milliseconds) and the standard errors in parentheses for the pandemic surveillance.

Algorithm	**SRAS**	**NAS**	**FULL**
Computational time	0.10 (0.01)	0.02 (0.01)	0.09 (0.02)

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